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INTERNSHIP REPORT ON THE PROJECT REVIEW AND TRACKING SYSTEM AT RWANDA WATER RESOURCES BOARD (RWB)

FACULTY: Information Technology

DEPARTMENT: Software Engineering

SUBMITTED BY: NDIKUMANA Norbert

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INTERNSHIP PERIOD: 10th August - 24th October, 2026

SUPERVISOR: NDAYISENGA Gilbert

DECLARATION

I, NDIKUMANA Norbert, declare that this internship report entitled “Project Review and Tracking System at Rwanda Water Resources Board” is my original work carried out at the Rwanda Water Resources Board (RWB). This report has not been submitted elsewhere for any academic qualification.

This report is the result of the practical training, research, and hands-on experience I gained during the internship period, and it accurately reflects the tasks, responsibilities, and knowledge acquired throughout my engagement. All sources of information used in the preparation of this report have been duly acknowledged.

Signature: .....

Date: .....

APPROVAL

I, NDAYISENGA Gilbert, hereby certify that the internship report entitled “Project Review and Tracking System at Rwanda Water Resources Board” carried out by NDIKUMANA Norbert has been approved for submission.

Signature: .....

Date: .....

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the Rwanda Water Resources Board (RWB) for offering me the opportunity to conduct my internship and contribute to the development of the Project Review and Tracking System. I am deeply thankful to my supervisor for the guidance, mentorship, and support extended to me throughout the internship period.

My heartfelt appreciation goes to my supervisor, whose guidance, mentorship, and continuous support played a critical role in the successful completion of my internship project. Their constructive feedback, encouragement, and availability whenever I faced challenges greatly enhanced my learning journey.

I also extend my sincere thanks to the entire team at RWB, who welcomed me warmly and created a collaborative and supportive atmosphere. Their teamwork, professionalism, and willingness to share knowledge made this internship not only productive but also enjoyable.

Finally, I am grateful to the Adventist University of Central Africa (AUCA) for facilitating this internship program and equipping me with the foundational skills needed to contribute meaningfully to such a project. This experience has been a cornerstone in my career development, and I remain committed to applying the lessons learned as I progress in the field of Software Engineering.

May God bless you all!

ABBREVIATIONS

RWB - Rwanda Water Resources Board
PRTS - Project Review and Tracking System
AUCA - Adventist University of Central Africa
MINEE - Ministry of Infrastructure and Environment
API - Application Programming Interface
REST - Representational State Transfer
RBAC - Role-Based Access Control
JWT - JSON Web Token
PDF - Portable Document Format
UI/UX - User Interface / User Experience
DB - Database
DM - Division Manager
MFA - Multi-Factor Authentication
WCAG - Web Content Accessibility Guidelines
SPA - Single Page Application

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CHAPTER I: GENERAL INTRODUCTION

1.1 Introduction

This internship was undertaken at the Rwanda Water Resources Board (RWB), the national government institution responsible for the management, protection, and sustainable development of Rwanda's water resources. RWB oversees water resource projects across all provinces - from dam construction and water treatment plants to irrigation schemes and flood mitigation - ensuring that every project submitted by private companies and organizations undergoes rigorous technical review before approval.

To fulfil this mandate efficiently, RWB needed a modern, digital platform to replace its legacy paper-based project submission and review processes. Before the system was built, external companies submitted project proposals through physical documents and email, Division Managers manually tracked review assignments on spreadsheets, and Reviewers had no centralized workspace for evaluating submissions. The result was traceability gaps, inconsistent review standards, delayed decision-making, and no real-time visibility of project status.

During the internship, I contributed to the Project Review and Tracking System (PRTS) - a secure, full-stack web application that digitizes the complete lifecycle of water-resource project management in Rwanda. PRTS connects external companies, Division Managers, Reviewers, Super Reviewers, and System Administrators in one governed platform, from the moment an external company registers and submits a project through the multi-stage review pipeline to the final approval or rejection decision.

Technically, PRTS is built on a Spring Boot backend with PostgreSQL persistence managed through Flyway migrations, a React frontend using Tailwind CSS and Framer Motion for responsive, animated interfaces, and a JWT-based authentication pipeline with multi-factor authentication (MFA) and fine-grained role-based access control. The system supports multi-language interfaces (English, Kinyarwanda, French), WCAG AA accessibility compliance, and branded PDF document generation. Working on this platform allowed me to strengthen my skills in enterprise backend development, relational database design, secure frontend development, state-machine workflow modelling, and application security, while developing professional soft skills and adapting to the institutional processes of a national government board.

This introductory chapter sets the foundation for the report by presenting the context of the internship, the host institution, and the focus of the project undertaken. The subsequent chapters provide a detailed overview of the organization, the tasks performed, the tools and technologies used, the challenges encountered, and the lessons learned throughout the internship period.

1.2 Objectives of the Internship

The main objectives of the internship were:

1. **To apply academic knowledge in a real-world software development environment:** The internship aimed to bridge the gap between theory and practice by applying concepts from software engineering, database management, and system security to an operational enterprise platform.
2. **To contribute to a secure, governed project review platform:** A core objective was to support a platform that digitizes the full lifecycle of water-resource project management - from company registration and project submission, through multi-stage reviewer assignment and recommendation, to the Division Manager's final approve or reject decision.
3. **To implement a multi-stage review workflow with state-machine enforcement:** Another objective was to help build workflow functionality that enforces the approval chain across External Companies, Division Managers, and Reviewers, with every state transition recorded, timestamped, and attributed to a named actor.
4. **To support role-based access control with permission-level granularity:** The internship involved contributing to a permission matrix that maps five distinct roles to twelve fine-grained permissions, enforced at both the API and UI levels.
5. **To strengthen application security and confidentiality:** Given the sensitivity of government infrastructure project data, a major focus was contributing to JWT-based authentication, multi-factor verification, and role-based access control.
6. **To develop professional and interpersonal skills:** Beyond technical competencies, the internship aimed to enhance communication, teamwork, and problem-solving through close collaboration with supervisors and colleagues.

1.3 Significance of the Internship

This internship was highly significant as it provided a valuable opportunity to apply academic knowledge in a real-world government water-resource management environment at RWB. Working on PRTS enabled me to contribute directly to a platform that improves the transparency, accountability, and efficiency of project review processes, and to observe how governed digital workflows strengthen institutional accountability and public trust.

Throughout the internship, I developed both technical and professional competencies. On the technical side, I gained hands-on experience in enterprise full-stack development with Spring Boot and React, relational database design with PostgreSQL and Flyway, state-machine workflow modelling, multi-factor authentication, and role-based access control. Professionally, I strengthened my teamwork, problem-solving, and communication skills and adapted to structured government workflows. Overall, the experience bridged academic learning with practical industry application and prepared me for a future career in enterprise software engineering and secure digital transformation.

CHAPTER II: OVERVIEW OF THE HOST ORGANIZATION

2.1 Historical Background

The Rwanda Water Resources Board (RWB) is a government institution established under Rwandan law with the mandate to manage, protect, and sustainably develop Rwanda's water resources. Operating under the supervision of the Ministry of Infrastructure, RWB is responsible for coordinating water resource projects across the country, including dam construction, water treatment and distribution systems, irrigation infrastructure, flood mitigation, and environmental protection of water bodies.

RWB was created as part of the Government of Rwanda's broader strategy to professionalize and centralize the management of natural resources, ensuring that water infrastructure projects are planned, reviewed, and executed according to national standards and international best practices. Prior to the establishment of RWB, water resource project oversight was distributed across multiple agencies, leading to inconsistent review processes, fragmented data management, and limited coordination between national and district-level planning.

Over the years, RWB has progressively embraced modern information and communication technologies to enhance service delivery and administrative efficiency. The Authority has deployed digital platforms for project submission, review tracking, and stakeholder communication, replacing manual paper-based processes that suffered from traceability gaps, inconsistent document formats, and slow multi-step approvals.

Today, RWB operates with headquarters in Kigali and a network of provincial and district offices across the country. It is led by a Board and Executive Director, and continues to play a vital role in financing national water infrastructure development, promoting sustainable water resource management, and supporting Rwanda's long-term environmental and economic transformation.

2.2 Mission & Vision

Mission

To manage and protect Rwanda's water resources sustainably through efficient, transparent, and evidence-based governance, ensuring equitable access to safe water and supporting national socio-economic development.

Vision

To become a world-class water resource management institution that ensures sustainable, equitable, and efficient water governance for Rwanda's prosperity.

Core Values

RWB's work is guided by a set of core institutional values, namely: Integrity, Transparency, Professionalism, Accountability, and Collaboration. These values shape the institution's culture and underpin its commitment to fair, efficient, and accountable water resource management.

2.3 Objectives and Mandate

RWB operates under a clear legal mandate, with the primary objective of managing and protecting Rwanda's water resources in support of national development. The key objectives and mandates include:

- **Water resource project oversight:** Reviewing and approving water resource project proposals submitted by private companies, NGOs, and government agencies to ensure compliance with national standards and environmental regulations.
- **Infrastructure planning and coordination:** Coordinating the planning, design, and implementation of water infrastructure projects - including dams, treatment plants, irrigation systems, and flood control - across all provinces.
- **Environmental protection:** Monitoring and enforcing environmental standards for water resource projects to protect Rwanda's rivers, lakes, and groundwater systems.
- **Stakeholder engagement:** Facilitating collaboration between government agencies, private sector companies, local communities, and international partners in the planning and execution of water resource projects.
- **Data management and reporting:** Maintaining a centralized repository of all water resource projects, their review decisions, and implementation status, and providing regular reports to the Ministry of Infrastructure and other stakeholders.
- **Modernizing project management:** Developing and managing secure digital systems that improve administrative efficiency, transparency, and accountability in the project review and approval process.

2.4 The Water Resources Management Division

The Water Resources Management Division is the core operational unit within RWB responsible for receiving, reviewing, and deciding on water resource project proposals. Its core responsibilities include the intake and registration of project submissions from external companies, the assignment of technical reviewers to submitted projects, the coordination of internal and final review decisions, and the maintenance of a complete audit trail of all review activities.

The Division is led, at the executive level, by the Director of Water Resources, and its work is carried out by Division Managers who assign projects to Reviewers, and by Reviewers who evaluate technical submissions. Super Reviewers provide read-only oversight, and System Administrators manage users, roles, and system configuration. Because the Division handles sensitive infrastructure project data - including financial projections, environmental impact assessments, and organizational details - its work demands strict confidentiality, careful audit trails, and tightly controlled access. It was within this context that the PRTS project was undertaken - to replace fragmented, paper-based processes with a governed, auditable, and scalable digital platform, which is the focus of the following chapter.

CHAPTER III: TASK DESCRIPTION AND INTERNSHIP EXPERIENCE

3.1 Roles and Responsibilities

During my internship at the Rwanda Water Resources Board (RWB), I worked on the development and enhancement of the Project Review and Tracking System (PRTS). My work spanned the full stack - backend services, database, frontend, and document generation - with particular focus on the project review workflow, role-based access control, multi-column filtering, and the user management interface.

On the backend, I contributed to Spring Boot controllers and services that implement the project lifecycle, worked with the state-machine model that tracks each project through its discrete statuses (DRAFT, SUBMITTED, IN_REVIEW, INFO_REQUESTED, APPROVED, REJECTED, RESUBMITTED, ARCHIVED), and helped enforce role-based access control at the API level. On the database side, I worked with the PostgreSQL schema and Flyway migrations that version the database and make deployments reproducible. On the frontend, I contributed to React and Tailwind CSS screens - dashboards, project tables, detail pages, and user management forms - ensuring responsive, animated, and consistent rendering.

Additional responsibilities included:

- Participating in requirement analysis and design discussions with supervisors and senior developers.
- Implementing and refining role-based dashboards for five distinct user roles (System Administrator, Division Manager, Reviewer, Super Reviewer, External User).
- Contributing to the multi-stage review workflow, including the reviewer assignment pipeline, internal recommendation, and global decision flow.
- Building per-column filter controls for the shared project table, enabling users to filter by any column with AND-combined logic.
- Designing and implementing the user creation modal with automatic role-based permission selection and additional permission overrides.
- Enforcing workflow integrity rules - ensuring Division Managers cannot make final rejection decisions in ASSIGN mode before a reviewer has provided an internal decision.
- Adding multi-language support (English, Kinyarwanda, French) across all user-facing screens.
- Implementing multi-factor authentication (MFA) with email-based verification codes.
- Writing and testing Flyway migrations and validating the database schema.
- Testing screens and workflows to ensure correctness at build time and run time.
- Preparing progress updates and presenting development milestones to supervisors.

These responsibilities gave me valuable hands-on experience in full-stack enterprise development, relational database administration, state-machine workflow design, and access-control enforcement, while strengthening my teamwork, communication, and professional problem-solving abilities.

3.2 Tools and Technologies Used

The following tools and technologies were used during the internship:

- **Java and Spring Boot:** for backend business logic, REST APIs, and the project review workflow.
- **Spring Security with JWT and MFA:** for stateless authentication, multi-factor verification, and API-level role-based access control enforced through permission checks.
- **PostgreSQL with Flyway:** for relational persistence and versioned, reproducible database schema migrations.
- **React with Tailwind CSS and Framer Motion:** for the component-based, responsive, and animated frontend.
- **Vite:** for frontend build tooling and hot module replacement during development.
- **Lucide React:** for the icon library used throughout the interface.
- **Chart.js / Recharts:** for dashboard analytics visualizations.
- **Git and GitHub:** for version control and team collaboration.
- **Maven:** for backend build automation and dependency management.

3.3 Tasks Completed

Over the course of the internship, I completed several tasks that contributed to the functionality, stability, and security of PRTS. I contributed to the project submission and review modules, helping ensure that external companies can register, submit projects with attachments, and track their progress through the review pipeline. I also worked on the multi-stage review workflow, helping track projects through their discrete states and supporting the assign, recommend, approve, and reject actions across the review chain.

I contributed to the per-column filtering system for the shared project table, building a reusable ColumnFilter component that enables users to filter any non-date column by distinct values, with AND-combined multi-column filtering and active filter chip display. I also worked on the user management interface, implementing automatic role-based permission selection during user creation, so that each role's default permissions are pre-selected while allowing administrators to grant additional permissions beyond the role defaults.

Additional completed tasks included:

- Implementing and refining role-based dashboard screens with analytics, monthly submission charts, and review completion trends.

- Enforcing workflow integrity - preventing Division Managers from bypassing the reviewer's internal decision with final rejection.
- Contributing to the global decision queue that shows projects awaiting Division Manager action after internal reviewer decisions.
- Implementing the registration approval workflow for external companies.
- Building the forgot-password flow with time-limited email codes and secure password reset.
- Helping enforce role-based access control and confidentiality controls across the API.
- Conducting build-time and run-time testing and synchronizing work through Git/GitHub.

Each of these tasks contributed to a governed, auditable, and standardized platform that supports RWB's water resource project review operations.

3.4 Application Development

Project Review and Tracking System (PRTS) was developed as a secure, full-stack web application for managing, reviewing, and tracking water resource projects across Rwanda. It digitizes the complete lifecycle of water-resource project management - from external company registration and project submission through multi-stage review assignment, internal recommendation, and final Division Manager decision - in a governed, auditable, and role-segregated manner.

Architecturally, the system combines a Spring Boot backend with PostgreSQL persistence (managed through Flyway migrations), a React and Tailwind CSS frontend with Framer Motion animations, and a multi-factor authentication pipeline secured by JWT-based stateless authentication. Role-based access control is enforced at the API level through a permission matrix mapping five roles to twelve fine-grained permissions, while the frontend mirrors these restrictions to ensure consistent access control. The following screens illustrate the main modules of the system.

The Login and Multi-Factor Authentication Portal

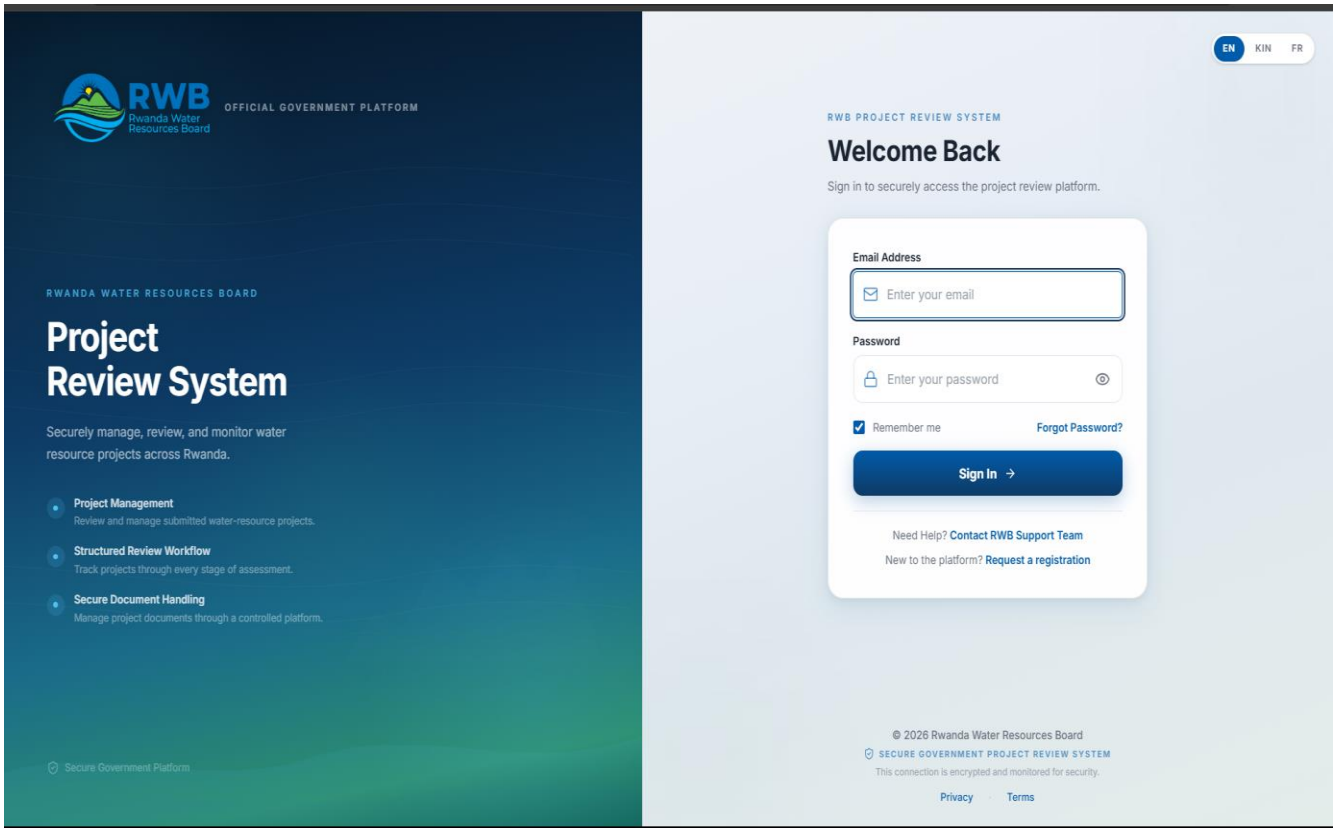


Figure 1: *The PRTS login and multi-factor authentication portal.*

The Login Portal is the secure entry point to PRTS for all users. It authenticates users through JWT-based stateless authentication and multi-factor verification before granting access to role-specific functionality.

- **JWT-Based Authentication:** Validates credentials and issues a signed JSON Web Token that authorizes each subsequent request without server-side sessions.
- **Multi-Factor Authentication:** Sends a 6-digit verification code via email (or returns it in dev mode) that must be entered before access is granted, adding a second layer of security.
- **Role Resolution:** On login, resolves the user's role and permissions so that the interface exposes only the actions that role is authorized to perform.
- **Forgot Password Flow:** Provides a time-limited email code for secure password reset, with brute-force protection (maximum five incorrect attempts before code invalidation).
- **Multi-Language Support:** The entire login flow is available in English, Kinyarwanda, and French.

The Role-Based Dashboard

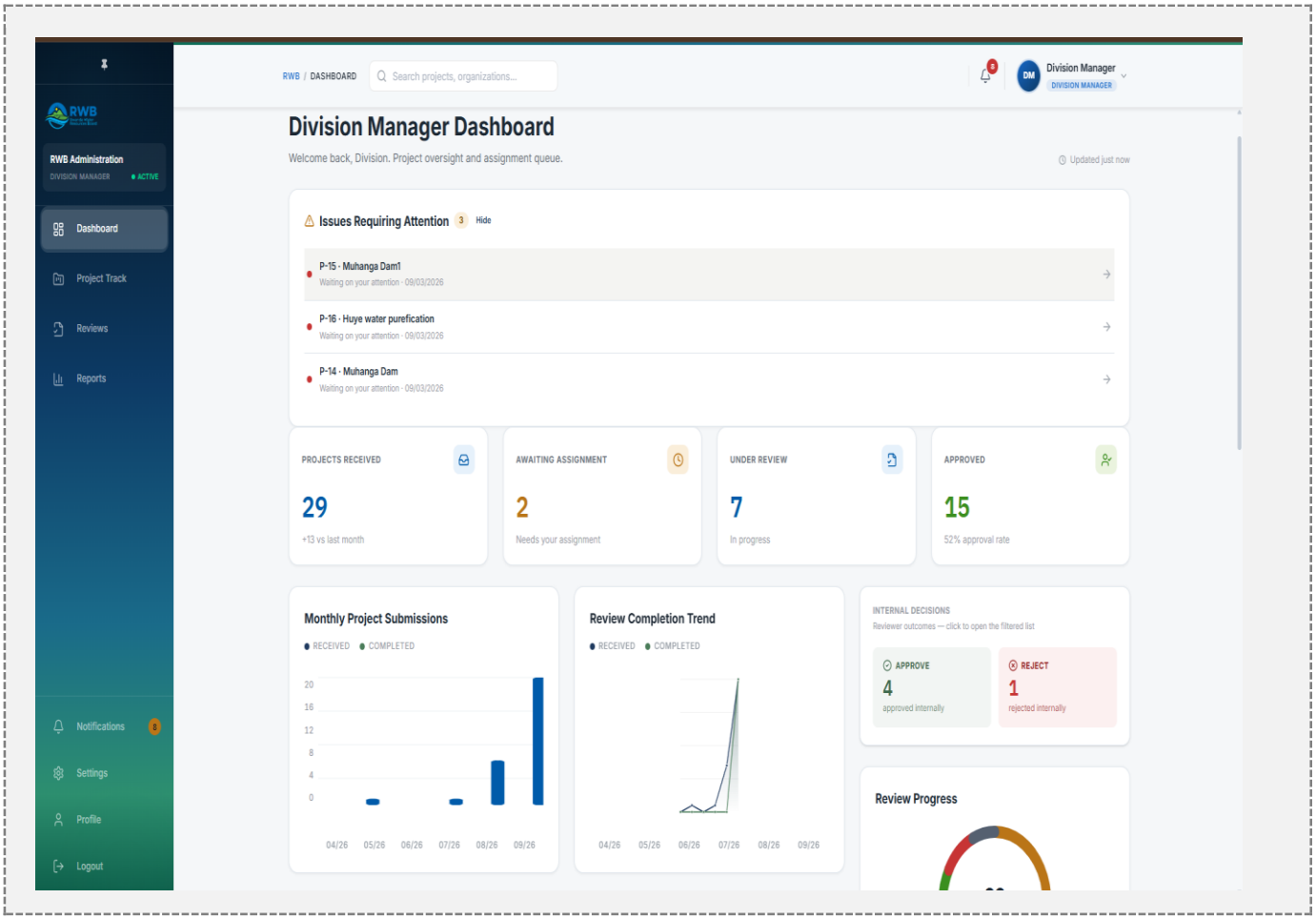


Figure 2: *Role-based dashboard showing project metrics and pending actions.*

After authentication, each user is presented with a role-based dashboard that shows only the projects, actions, and metrics relevant to their position.

- **Role-Specific View:** Presents System Administrators, Division Managers, Reviewers, Super Reviewers, and External Users with tailored dashboards and action sets.
- **Live Project Visibility:** Surfaces project status counts, pending approvals, and outstanding tasks, giving management real-time oversight of the Division's workload.
- **Analytics Charts:** Displays monthly project submission trends and review completion trends through interactive charts.
- **Issues Requiring Attention:** Surfaces overdue reviews, rejected projects needing follow-up, and SLA risks with direct navigation to the affected projects.
- **Quick Navigation:** Provides direct entry points to project submission, reviewer assignment, reports, and user management.

External Company Project Submission

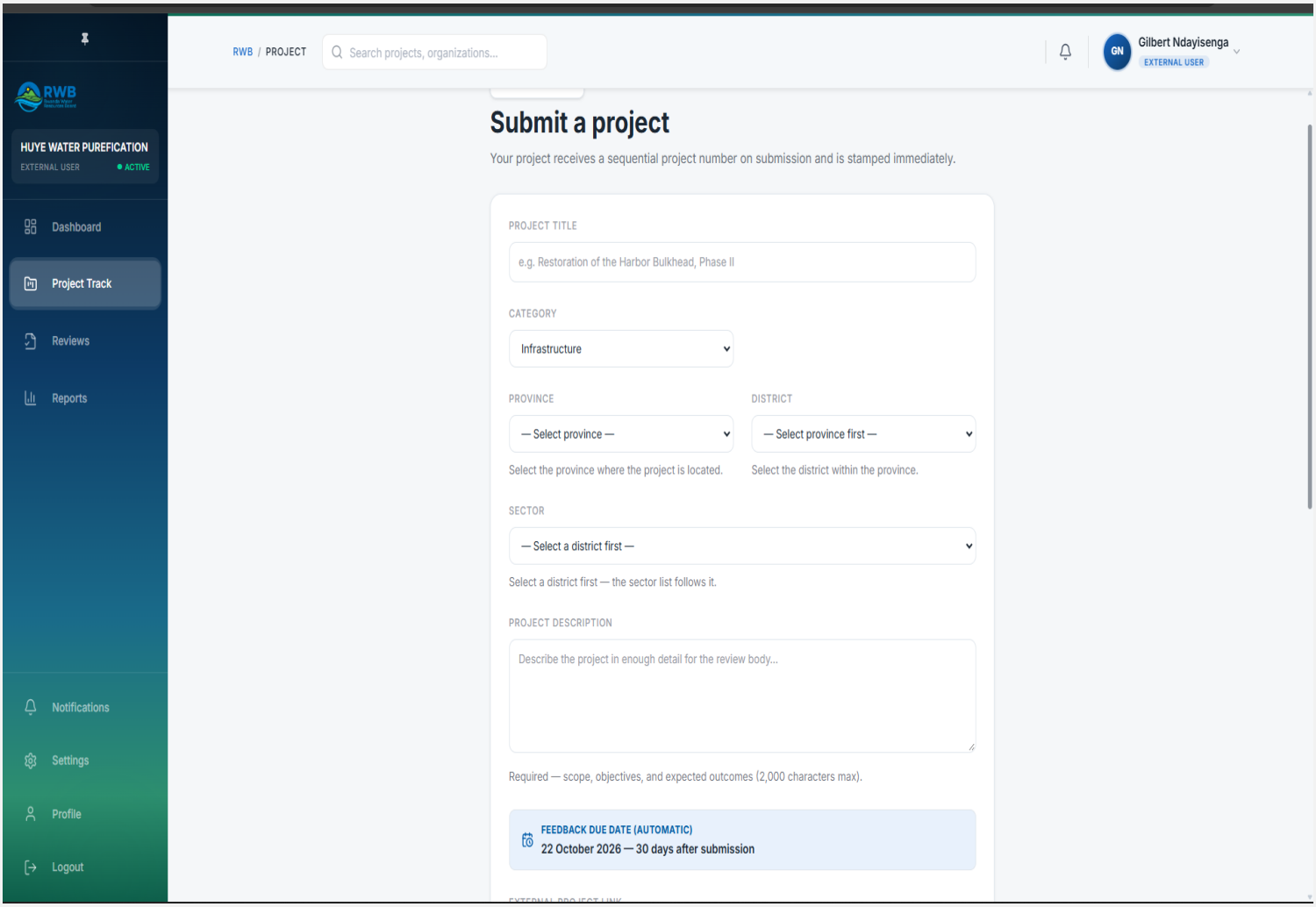


Figure 3: *External company project submission form with file uploads.*

The Project Submission screen is where an external company creates and submits a water resource project proposal.

- **Structured Classification:** Captures project name, category, province, district, sector, and detailed description.
- **File Attachments:** Supports multiple file attachments including documents, ZIP archives, and project links.
- **Status Tracking:** Projects begin in DRAFT status and transition to SUBMITTED when the company is satisfied.
- **Resubmission Support:** Rejected projects can be resubmitted after the company addresses review feedback.

Project Track with Per-Column Filters

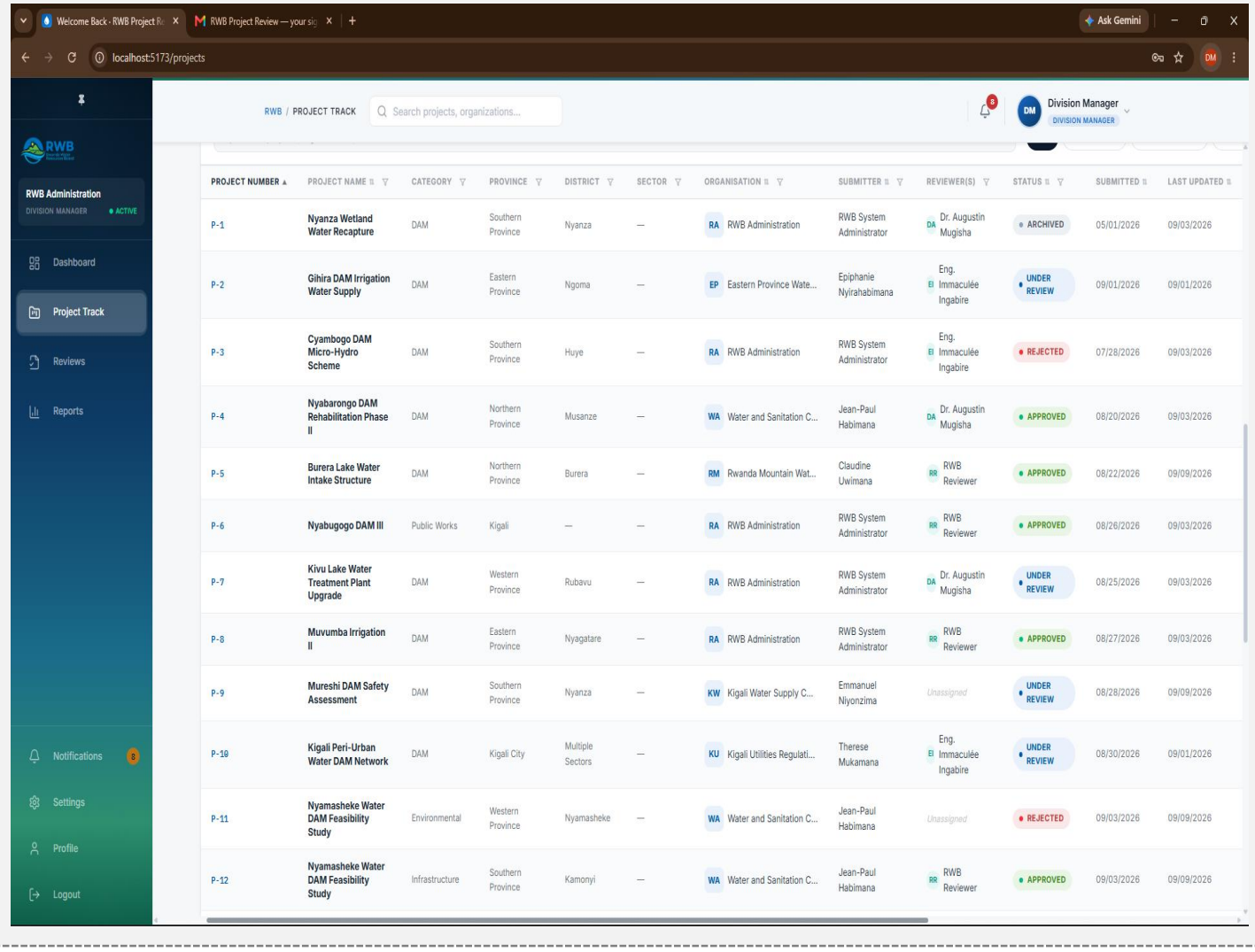


Figure 4: *Project Track page with per-column filter controls and status tabs.*

The Project Track page is the central monitoring interface where Division Managers and other authorized roles view and manage all projects in the system.

- **Per-Column Filter Controls:** Each non-date column header includes a filter icon that opens a dropdown listing all distinct values in that column.
- **Status Tabs:** Quick-access tabs for ALL, SUBMITTED, UNDER REVIEW, APPROVED, REJECTED, and ARCHIVED provide instant status filtering.
- **Header Sort Icons:** Clickable column headers enable ascending/descending sort.
- **Search Bar:** Real-time search across project ID, name, and organization name.
- **Action Buttons:** SUBMITTED projects show “Review” and “Assign” buttons.

Division Manager's Project Detail with Reviewer Assignment

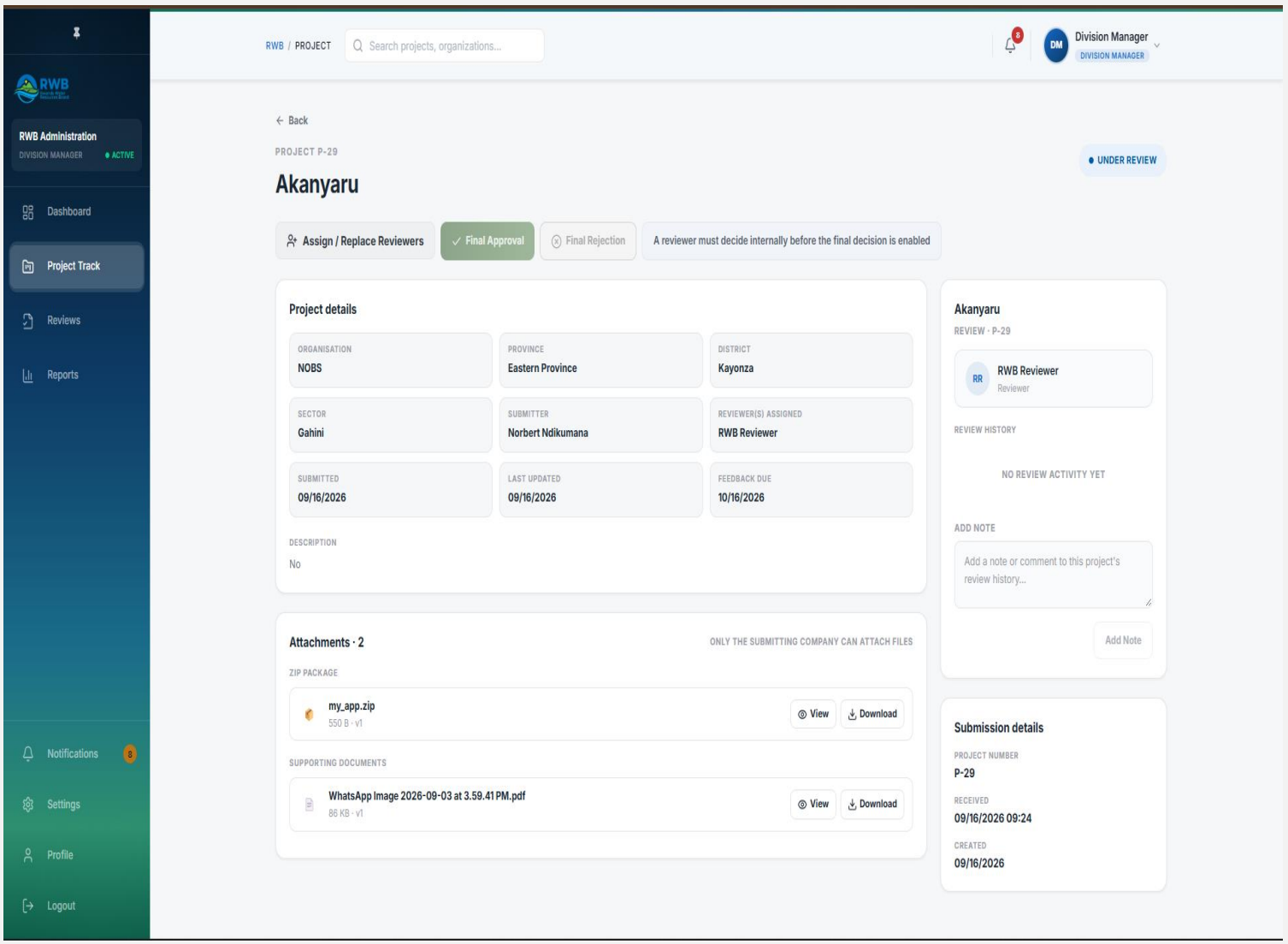


Figure 5: *Division Manager's project detail view with reviewer assignment.*

The Project Detail page shows the complete information for a single project, including its full review history, attachments, and available workflow actions.

- **Two Review Modes:** The Division Manager can choose REVIEW mode or ASSIGN mode.
- **Workflow Integrity Enforcement:** In ASSIGN mode, both the "Final Approval" and "Final Rejection" buttons are disabled until the assigned reviewer has provided an internal decision.
- **Review History:** Displays the complete decision history with timestamps, decision levels (INTERNAL vs GLOBAL), and officer attribution.
- **Notes and Comments:** Supports adding contextual notes to the review history.
- **Project Details Panel:** Shows organization, province, district, sector, submitter, assigned reviewers, submission date, and last update.

Reviewer's Personal Review Queue

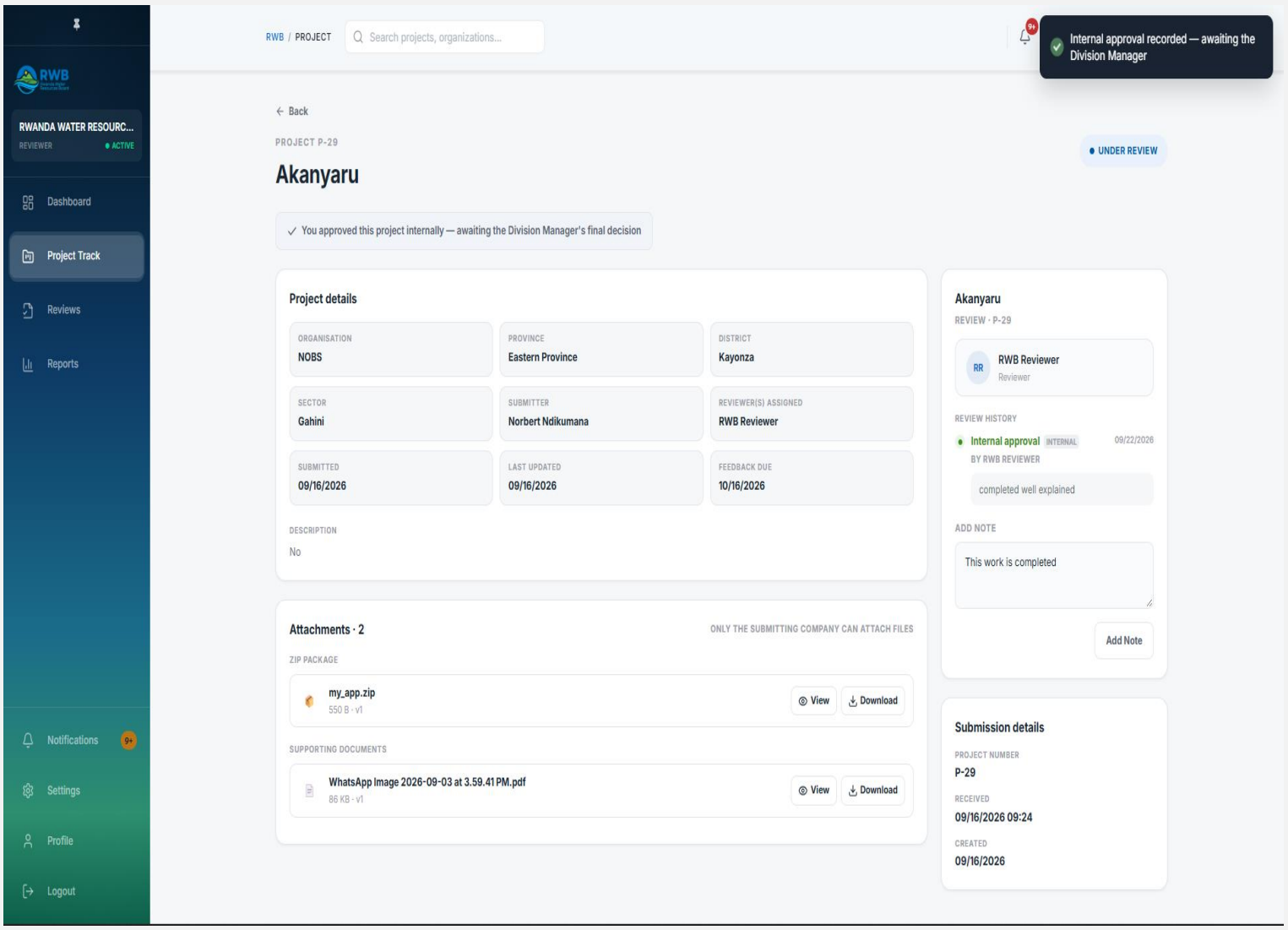


Figure 6: *Reviewer's personal review queue with recommendation actions.*

The Review Queue is the workspace where Reviewers access their assigned projects and submit internal recommendations.

- **Personal Queue:** Shows only projects assigned to the current reviewer.
- **Recommendation Actions:** The reviewer can submit Approve Internally, Request Information, or Reject Internally - each with mandatory reasoning.
- **Internal Decision Tracking:** All internal recommendations are recorded with the reviewer's identity, timestamp, and reasoning.
- **Statistics:** Displays the reviewer's completion statistics including total assigned, completed, and approval/rejection rates.

Multi-Stage Review Workflow

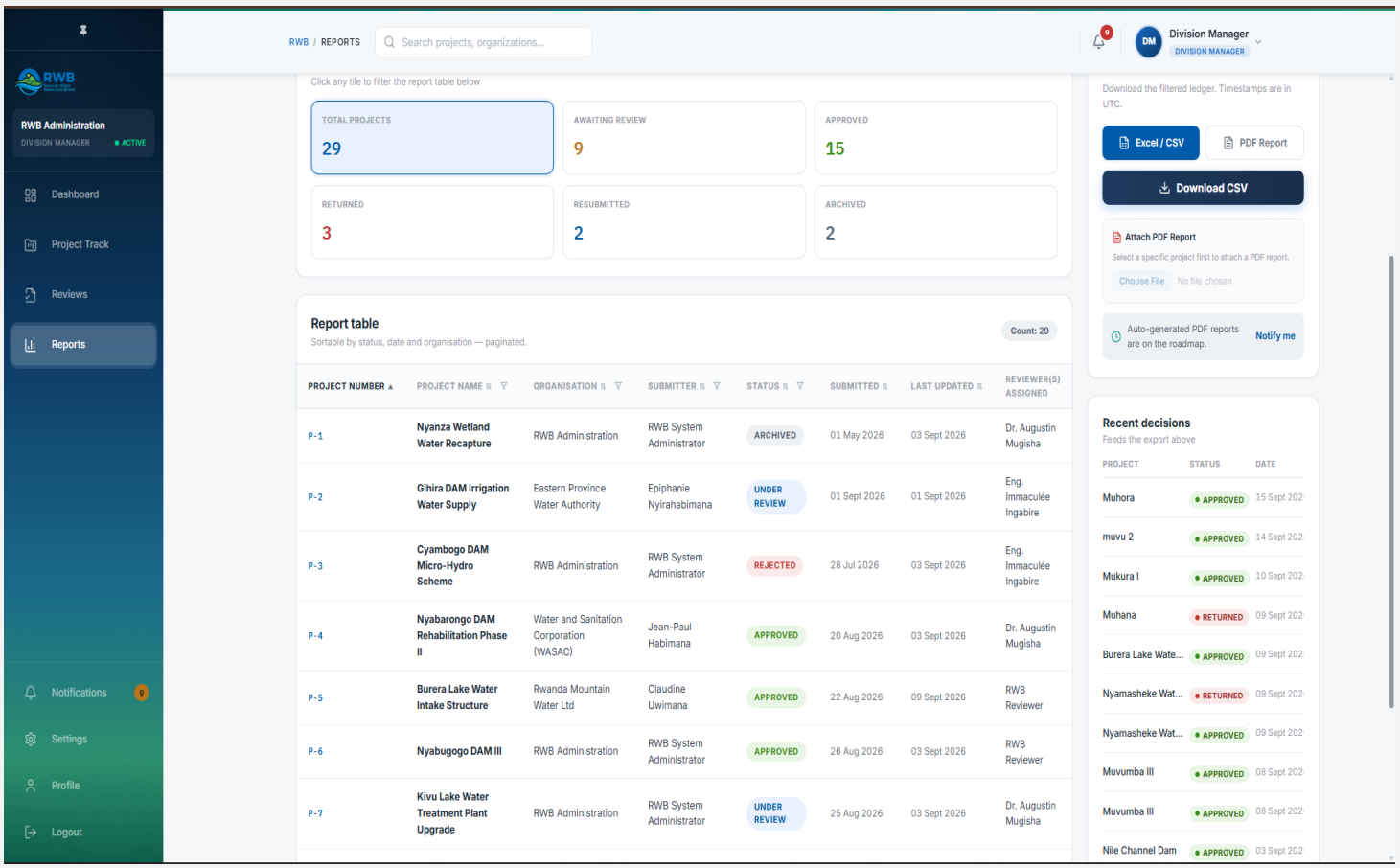


Figure 7: *Multi-stage review workflow with internal and global decision pipeline.*

The Review Workflow is the governed pipeline through which every project passes after submission.

- **State-Machine Enforcement:** The project status follows a strictly defined state machine with validated transitions.
- **Two-Tier Decision Model:** Projects in ASSIGN mode require an internal reviewer decision before the Division Manager can make the global final decision.
- **Rejection and Resubmission:** Rejected projects can be resubmitted, creating a feedback loop with full history.
- **Information Requests:** Either the Division Manager or Reviewer can request additional information from the external company.

Project Detail with Full Review History and Audit Trail

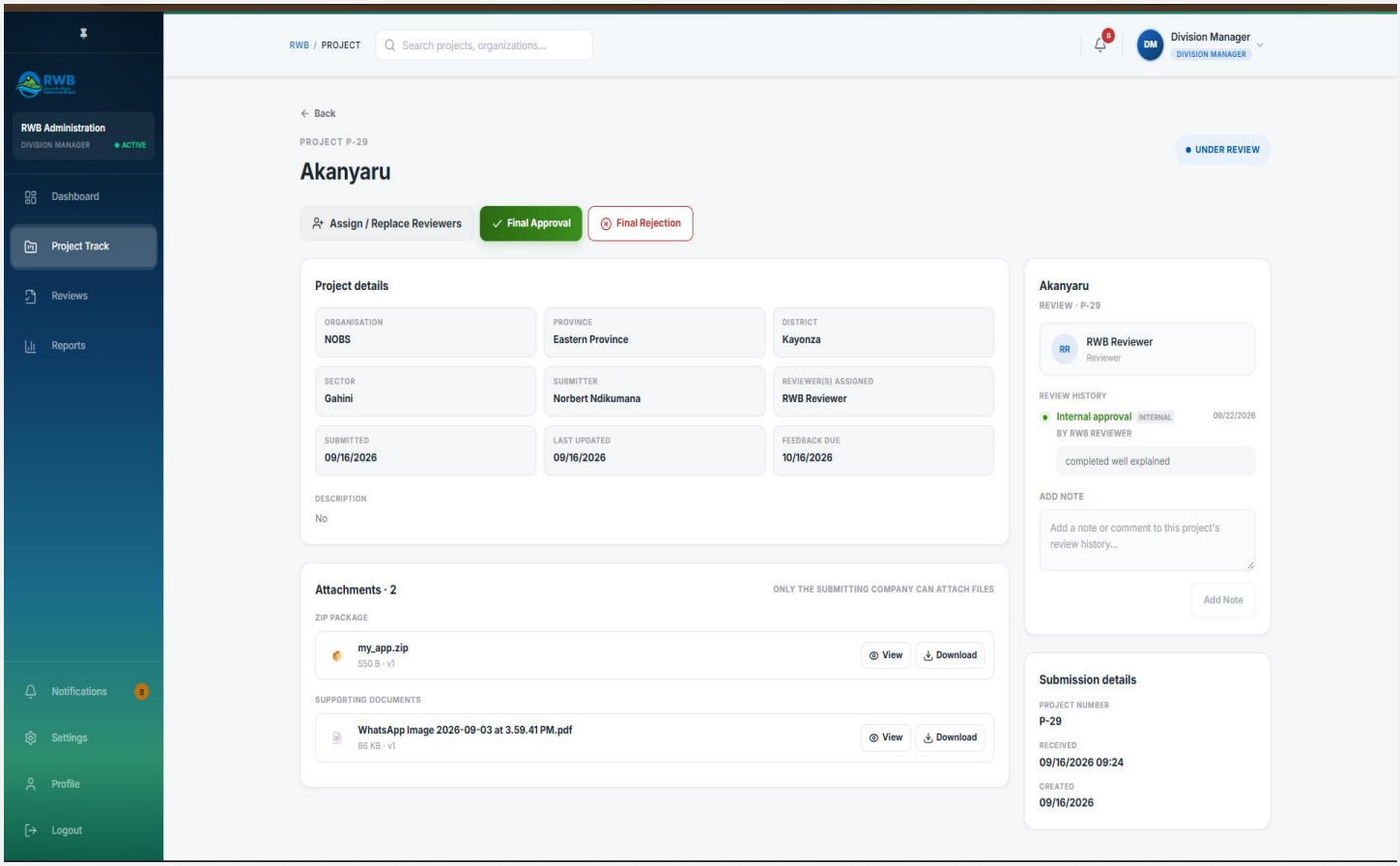


Figure 8: *Project detail page showing full review history and decision audit trail.*

Every project in PRTS maintains a complete, immutable audit trail of all decisions, notes, and status transitions.

- **Decision Level Display:** Each entry shows whether it was an INTERNAL or GLOBAL decision.
- **Officer Attribution:** Every decision is attributed to a named officer with role, timestamp, and reasoning.
- **Status Transitions:** The project status badge updates in real-time.
- **External Company View:** External companies can see the review history of their own projects.

System Administrator Dashboard

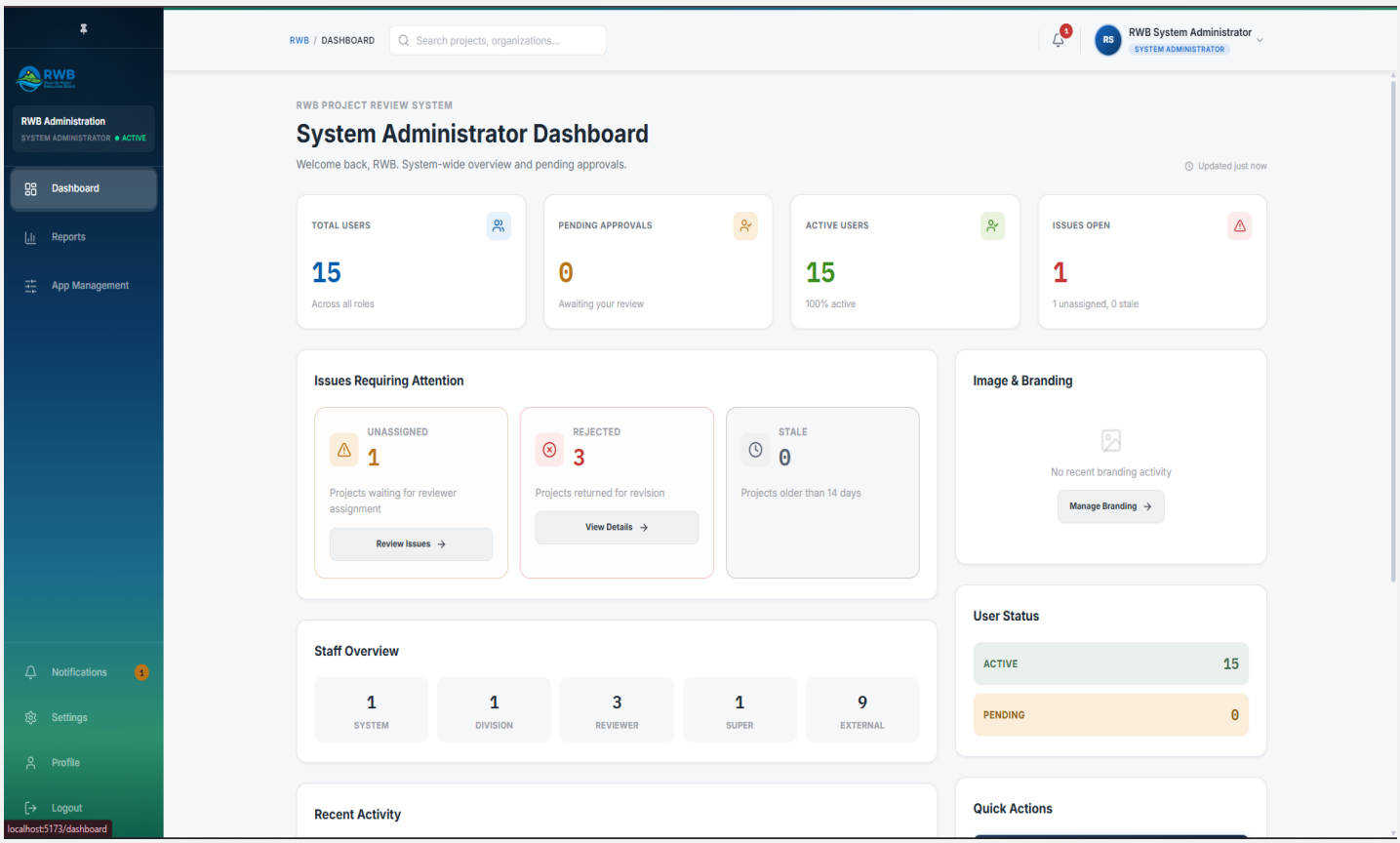


Figure 9: *System Administrator dashboard with analytics and activity feed.*

The System Administrator has access to a comprehensive dashboard that provides system-wide visibility.

- **Platform-Wide Analytics:** Displays total projects received, completed, and pending across all organizations and provinces.
- **Recent Activity Feed:** Shows a real-time log of system events with timestamps.
- **Organization Breakdown:** Lists all organizations with their project counts.
- **Issues Requiring Attention:** Surfaces system-wide issues including overdue reviews.
- **Quick Actions:** Provides direct links to user management, reports, and system configuration.

User Management with Role-Based Permission Preview

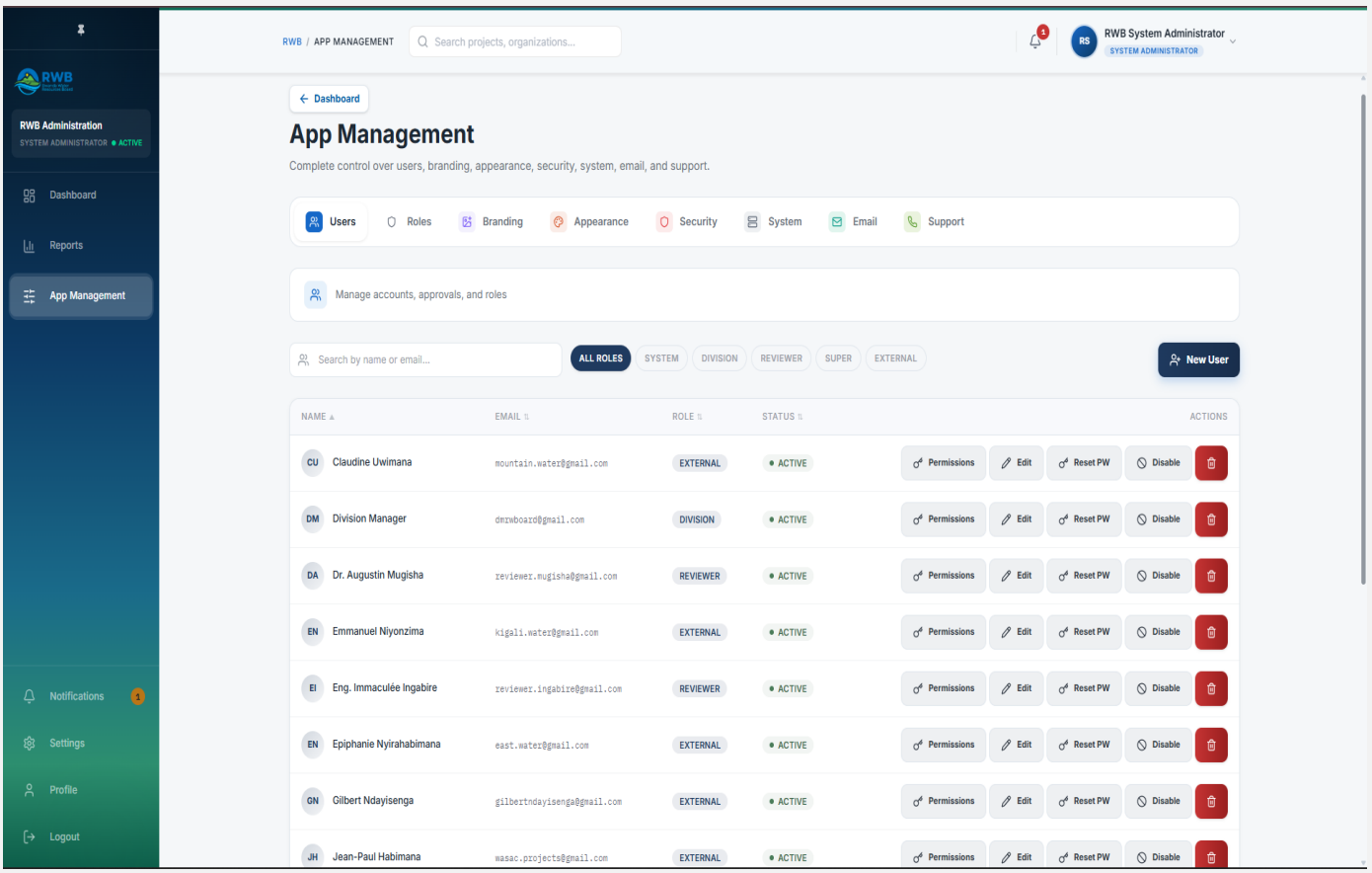


Figure 10: *User management screen with role assignment and permission preview.*

The User Management screen is the security-administration panel of PRTS.

- **Role-Based Auto-Selection:** When creating or editing a user, selecting a role automatically pre-checks that role's default permissions.
- **User Lifecycle Management:** Supports the complete account lifecycle from registration through admin approval to active status.
- **Permission Matrix:** Twelve fine-grained permissions mapped to five base roles, customizable per user.
- **Audit Trail:** All user management actions are logged with timestamps.

App Management Panel

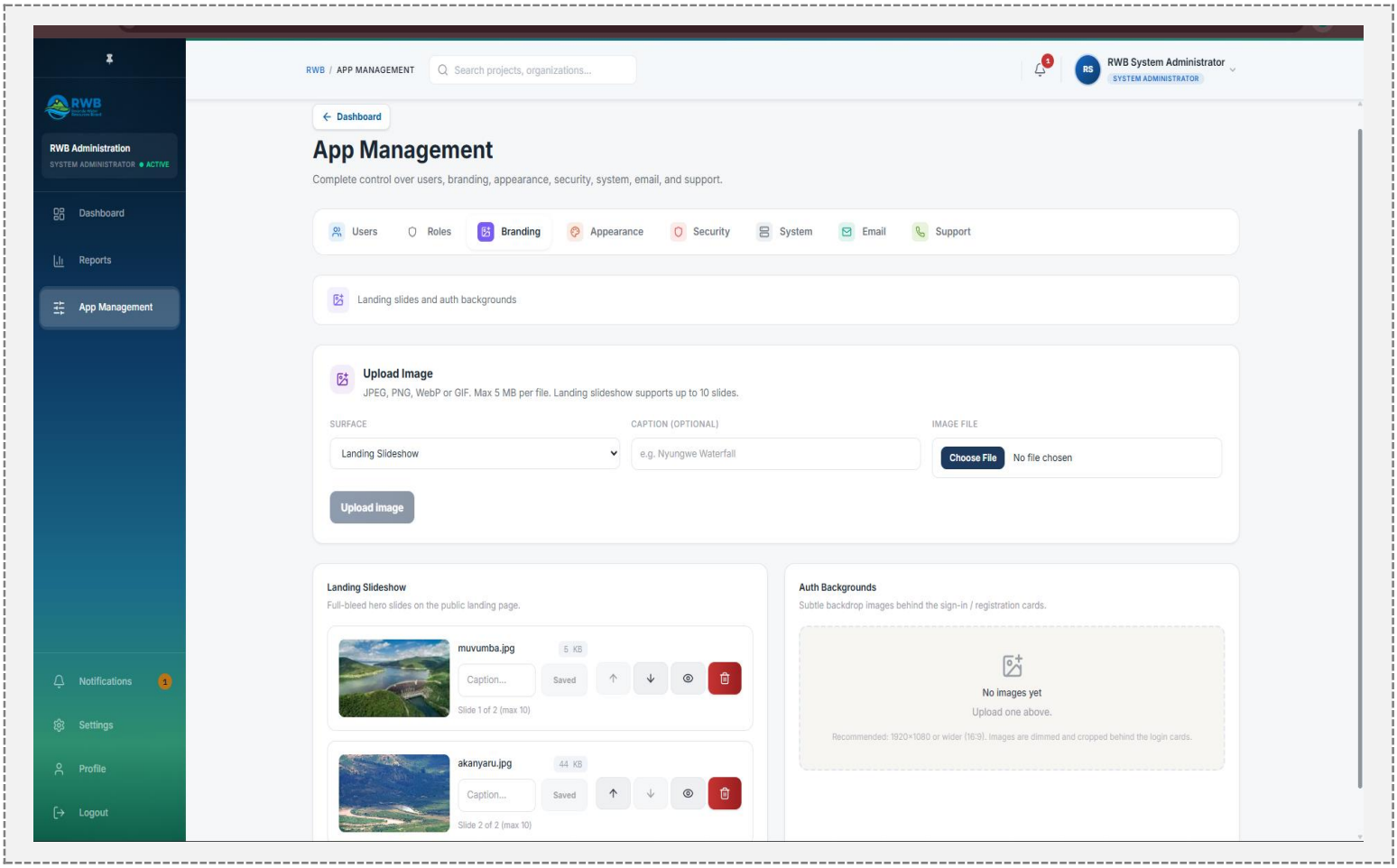
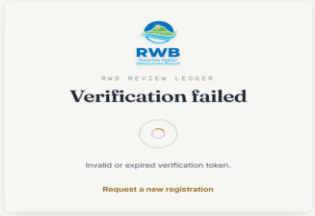


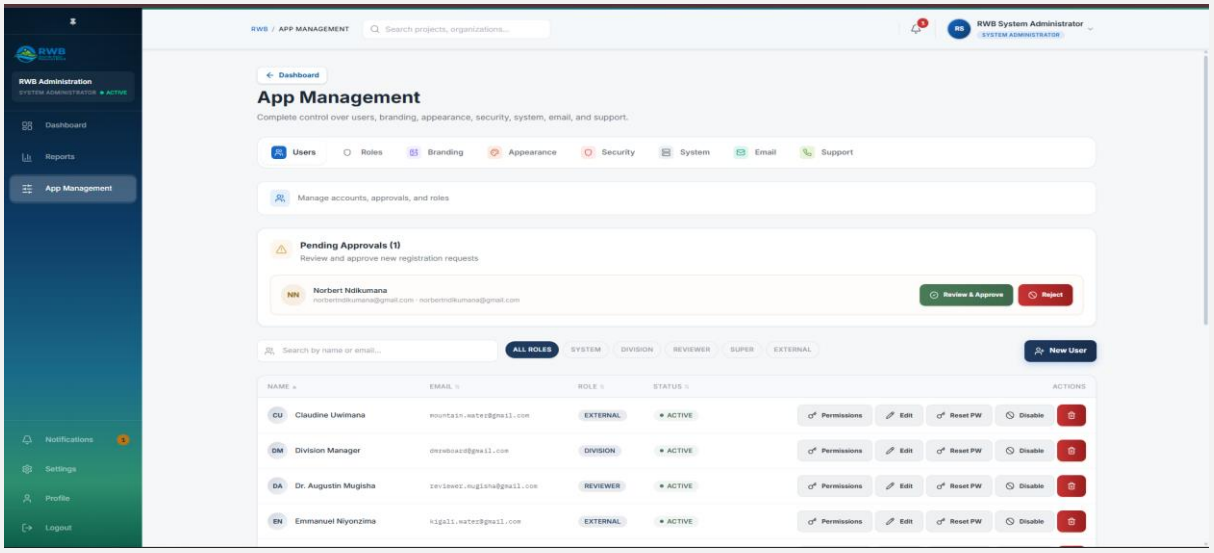
Figure 11: *App Management panel with appearance, security, and email settings.*

The App Management panel allows the System Administrator to configure system-wide settings.

- **Appearance Settings:** Manages landing page hero images and authentication page backgrounds.
- **Security Settings:** Configures password policies, MFA requirements, and session management.
- **Email Settings:** Manages SMTP configuration for MFA codes and password reset emails.
- **Feature Toggles:** Enables or disables optional modules.

Registration Approval Workflow





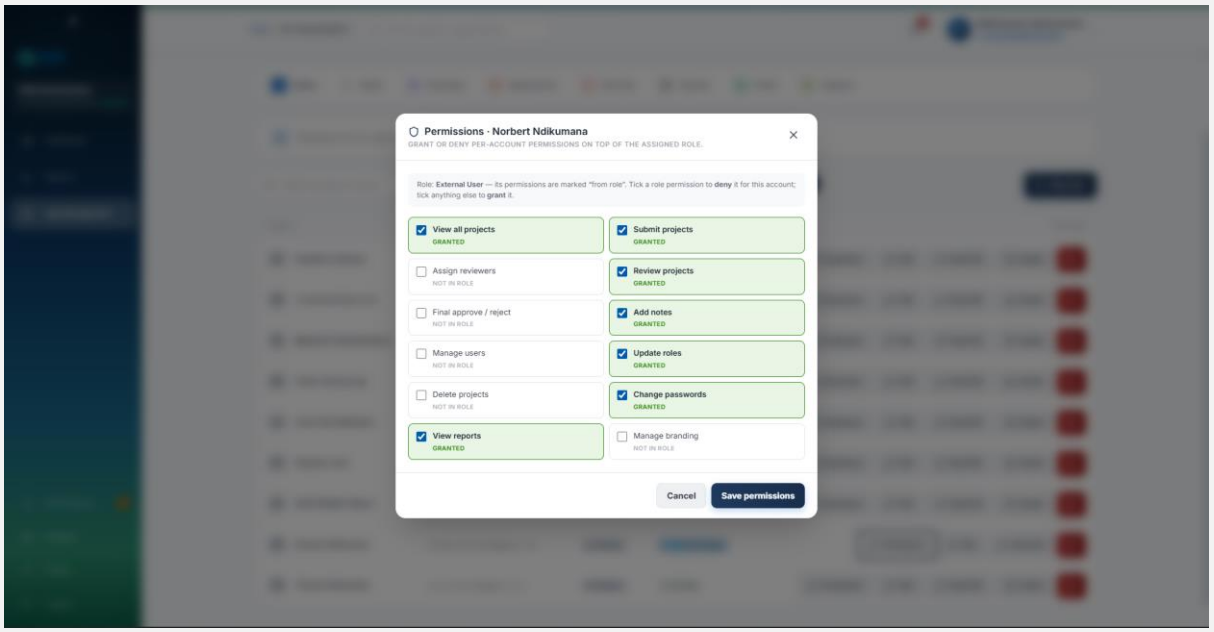


Figure 12: *Registration approval workflow and external company lifecycle.*

PRTS supports a complete registration and approval workflow for external companies.

- **Self-Registration:** External companies register through a public form.
- **Email Verification:** A verification email is sent for identity confirmation.
- **Admin Approval:** After email verification, the account awaits administrator approval.
- **First Project Requirement:** Newly approved users are encouraged to submit immediately.
- **Full Access:** After submitting their first project, the account transitions to full active status.

CHAPTER IV: CHALLENGES AND CONCLUSION

4.1 Challenges Faced

During the internship, several challenges were encountered while developing and stabilizing the Project Review and Tracking System (PRTS). These can be grouped into workflow, frontend, security, and deployment challenges:

1. **Modelling a complex, multi-state workflow:** The project lifecycle spans eight discrete states (DRAFT, SUBMITTED, IN_REVIEW, INFO_REQUESTED, APPROVED, REJECTED, RESUBMITTED, ARCHIVED) across several roles, with two-tier decision requirements in ASSIGN mode.
2. **Building reusable, filterable data tables:** Creating a shared table component with per-column filter controls - where each column header has a dropdown listing distinct values, multiple column filters combine with AND logic, and active filters display as clearable chips - required careful abstraction.
3. **Enforcing fine-grained role-based access control:** With five roles and twelve distinct permissions, ensuring that each endpoint and UI element was protected by the correct permission demanded careful, systematic testing.
4. **Implementing workflow integrity guards:** A critical challenge was ensuring that Division Managers in ASSIGN mode could not bypass the reviewer's internal decision. The initial implementation allowed final rejection without any internal review, which violated the intended workflow.
5. **Managing reproducible database migrations:** Keeping the PostgreSQL schema versioned and reproducible through Flyway, and handling migration numbering and repair cleanly, was a recurring practical challenge.

4.2 Lessons Learnt

The internship was a valuable learning experience that contributed significantly to my technical and professional development. Key lessons learned include:

1. **Enterprise backend development with Spring Boot:** I gained practical experience building REST APIs, services, and business logic, and learned how a well-structured backend supports a complex, governed workflow.
2. **Workflow and state-machine design:** Working with an eight-state project lifecycle taught me the importance of modelling state transitions carefully and enforcing transition guards both server-side and client-side.
3. **Reusable component design:** Building the shared ColumnFilter component taught me how to design flexible, reusable UI components that adapt to different data shapes and column configurations.
4. **Application security and RBAC:** Implementing JWT authentication, multi-factor verification, and fine-grained role-based access control deepened my understanding of securing sensitive systems.

5. **Database migration management:** Working with PostgreSQL and Flyway taught me how to version schemas and make deployments reproducible.
6. **Teamwork and version control:** Through collaborative work and the use of Git and GitHub, I gained practical experience in team-based software engineering workflows.

4.3 Conclusion

The internship at the Rwanda Water Resources Board (RWB), working on the Project Review and Tracking System (PRTS), was a highly valuable experience that bridged the gap between theoretical knowledge and practical application. Through my involvement in the development of PRTS, I applied academic concepts in system analysis, database design, backend and frontend development, state-machine workflow modelling, multi-factor authentication, and application security, while strengthening my problem-solving, teamwork, and professional communication skills.

This internship served as a vital platform for technical and professional growth, providing hands-on experience in secure enterprise systems and collaborative development practices. By contributing to PRTS, I developed critical skills in software engineering and system security that establish a solid foundation for my career and will enable me to build secure, reliable digital solutions for government institutions and beyond.